

Difficulty and Party Affiliation for Voters Turnout

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1 Importance and Context

High voter turnout is crucial for ensuring fairness and equal representation of all parties in the United States Presidential Election. According to research by the Pew Research Center, only 37% of eligible voters participated in all three of the 2018, 2020, and 2022 elections, while 30% did not vote in any of these elections. Voter turnout was 49% in the 2018 midterm elections and increased to 66% in the 2020 presidential election. Despite this 17% increase, work still needs to be done to ensure that the voices of the remaining 34% of eligible voters are heard and represented.

Furthermore, among those who voted in both the 2018 and 2020 elections, 55% were Democrats and 40% were Republicans. However, in the group that voted in both the 2020 and 2022 elections, 57% were Republicans and 40% were Democrats. This consistent difference of about 15% in party representation raises questions about whether voters from different parties experience varying levels of difficulty when voting, which might contribute to the disparity in turnout between parties.

To help improve voter turnout rates, this analysis explores whether there is a difference in voting difficulty between Democratic and Republican voters. Specifically, it addresses the following research question:

Do Democratic voters or Republican voters experience more difficulty voting?

Answering this question can guide government officials in addressing the right problems when developing solutions to make voting easier for everyone. By focusing on the appropriate issues, efforts to improve the voting system can become more efficient. More importantly, the results of this analysis can contribute to more balanced turnout across political parties in presidential elections, thereby reducing party polarization and ensuring fairness for all eligible voters, regardless of their political affiliation.

2 Data and Methodology

This report investigates whether there is a difference in voting difficulty between Democratic and Republican voters. In our study, "Voting difficulty" is represented by the number of barriers or challenges a voter perceives when participating in an election and the overall perceived difficulty level of voting. This concept is analyzed by political affiliation (Democrats and Republicans) to compare their perceptions of voting difficulty.

In the study we leverage data from the American National Election Studies (ANES) 2022 Pilot Study. This dataset is a cross-sectional survey conducted to test new questions under consideration for potential inclusion in the ANES 2024 Time Series Study. It explores a wide range of topics, such as opinion on popular policy issues, voter behavior, and sentiments toward different demographics. This observational dataset is collected via the YouGov panel using non-probability sampling. This method produces a sample that looks similar to a probability sample on the match characteristics but may still differ in unknown ways on unmatched characteristics. It is important to note that the YouGov panel is not nationally representative and consists of participants who sign up to complete questionnaires in exchange for rewards.

This dataset includes responses from 1585 individuals. However, 85 cases were not selected by the sampling matching procedure (indicated by "NA" in the column "weight") and were subsequently excluded from our analysis. Additionally, we excluded individuals who did not register to vote ("No, not registered" in the column "reg"), as their opinions on voting difficulty may not be as informative and relevant since they did not engage in the registration process. We recognize that there are individuals who did not vote for 2022 presidential election turnout ("No, not registered", in the column "turnout22"), but we kept their responses because upon further investigation, all had responses in the voter difficulty section of the survey and selected at least one reason for voting difficulty.

Next, we categorized the observations into either the Republican group or the Democrat group. We used the "pid_x" column, which had the highest response rate from survey takers regarding their party affiliation. The question was on a 7-point Likert scale ranging from 1 ("Strong Democrat") to 7 ("Strong Republican"). We grouped respondents who identified as "Strong Democrat" to "Lean Democrat" into the Democrat group, denoted as 1, and those who identified as "Lean Republican" to "Strong Republican" into the Republican group, denoted as 2. Respondents who identified as "Independent" or skipped the question were excluded from further analysis. After these adjustments, our final sample consisted of 1,143 observations.

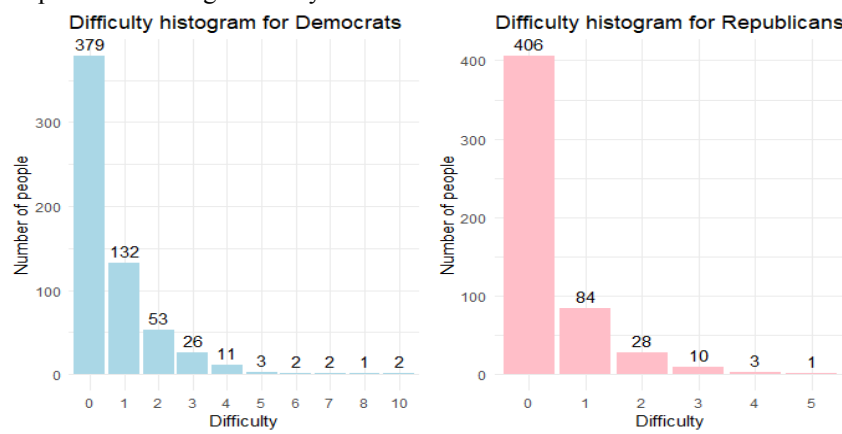
To answer our research question, we adopted two methods.

Method 1: We created the variable "overall_hard" based on the "Vharder" section in the survey, which included 11 questions asking respondents about the difficulty of different parts of the voting process. Respondents could respond with "1" if they found the aspect difficult and "2" if they did not. We converted all "2"s to "0"s, so the sum of a respondent's answers to the 11 questions ranged from 0 to 11, where 0 represents no perceived difficulty and 11 represents the highest perceived difficulty. This variable, "overall_hard," provided an intuitive scale to compare perceived voter difficulty for each observation.

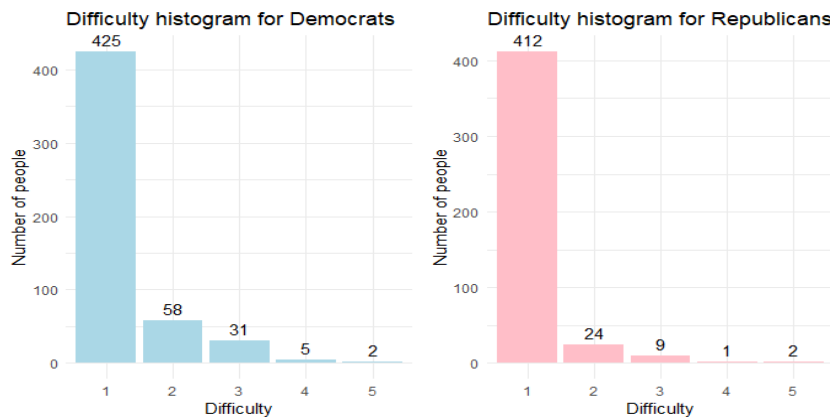
Method 2: We used the "votehard" column, which is based on an ordinal Likert scale question asking respondents to rate their overall difficulty in voting. The scale ranges from 1 ("not at all difficult") to 5 ("extremely difficult").

We plotted the number of respondents per difficulty level as defined in both Method 1 and Method 2 to visualize the distribution of perceived voting difficulty.

Method 1



Method 2



The **Null Hypothesis** for our test is phrased as such:

A member of the Democrat group experiences the same difficulty level in voting as a member of the Republican group.

Based on the data we are working with, we adopted the Wilcoxon Rank-Sum test as the statistical method. For a Wilcoxon Rank-Sum test to produce reliable inference, the following must be true: the data must be drawn from an i.i.d. sample and must be measured on at least an ordinal scale. We address each of these requirements in turn.

To begin with, the data must be generated through an i.i.d. process. The ANES 2020 pilot uses a panel of individuals from the YouGov platform, which introduces the possibility of dependencies. For example, there could be non-response bias, where certain types of individuals are more likely to respond to surveys, such as those looking to make extra money by taking surveys. This could result in potential bias in responses. However, since YouGov

claims to have millions of users, the large range of the population it covers may help mitigate non-response bias. Additionally, the differing political views of Democrats and Republicans help mitigate dependencies between the data from these two groups.

Last but not least, the outcome variable must be measured on at least an ordinal scale. We chose two variables to conduct the statistical test, so we will discuss them one by one.

In Method 1, the variable "overall_hard" is essentially a count of how many aspects of voting people find difficult, ranging from 0-11. A higher number indicates that an individual finds more areas of the voting process difficult, suggesting they find voting more difficult in general. We can argue that this is an ordinal variable because the difference in difficulty level between finding 1 area difficult versus 2 areas (e.g., "my work schedule" vs. both "my work schedule" and "bad weather") is not exactly quantified by discrete numbers. One can also argue that it is a metric value since it is a count of reasons, and the difference between 3 and 5 represents 2 reasons for voting difficulty. Since the Wilcoxon Rank-Sum test requires at least an ordinal variable, the overall_hard score is still eligible for the test.

In Method 2, we used the variable "votehard" from the survey. Votehard is a typical Likert scale used to evaluate people's opinions and perceptions of voting difficulty. Although the sentimental distance between consecutive levels in votehard is aimed to be kept roughly constant, it is not exactly quantified, making it an ordinal measure.

3 Results

```
# Method 1
wilcox_test_result_org <- wilcox.test(democrats$overall_hard, republicans$overall_hard)
print(wilcox_test_result_org)
# W = 187526, p-value = 3.998e-08
# Method 2
wilcox_test_result_samp_2 <- wilcox.test(democrats_alter$votehard, republicans_alter$votehard)
print(wilcox_test_result_samp_2)
# W = 128919, p-value = 2.351e-06
```

The Wilcoxon rank-sum tests using both Method 1 and Method 2 resulted in the same conclusion. Method 1 had a p-value of 3.998e-08, and Method 2 had a p-value of 3.351e-08. The results of the Wilcoxon rank-sum test indicate a statistically significant difference in voting difficulty between Democratic and Republican voters since both p-values are less than 0.05. This leads the team to reject the null hypothesis that there is no difference in voting difficulty between the two groups. These results suggest that Democratic voters perceive greater voting difficulties compared to Republican voters.

4 Discussion

This study found evidence that Democratic and Republican voters do not experience the same level of difficulties in voting. The difference appears practically significant, with Democrats experiencing a higher difficulty level than Republicans.

Results from our study may help government officials to identify systematic barriers in voting, and focus on developing solutions that help the right population to address voting difficulty that affects turnout. While this study addresses the difficulties in voting in general, we'd be interested to learn about what are the reasons and specific areas that citizens find difficult in the voting process, and if there is a difference in the reasons between Democrats and Republicans. Combined with our study, government officials and politicians would have a holistic understanding of voting difficulty that the population faces, be able to find the right obstacle to tackle, which will ensure fairness to all citizens in the presidential election, encourage a more equal turnout rate in all political parties, and further strengthen the democratic process in the United States.